From acceleration to velocity

When acceleration stays the same, velocity changes by the same amount each second.

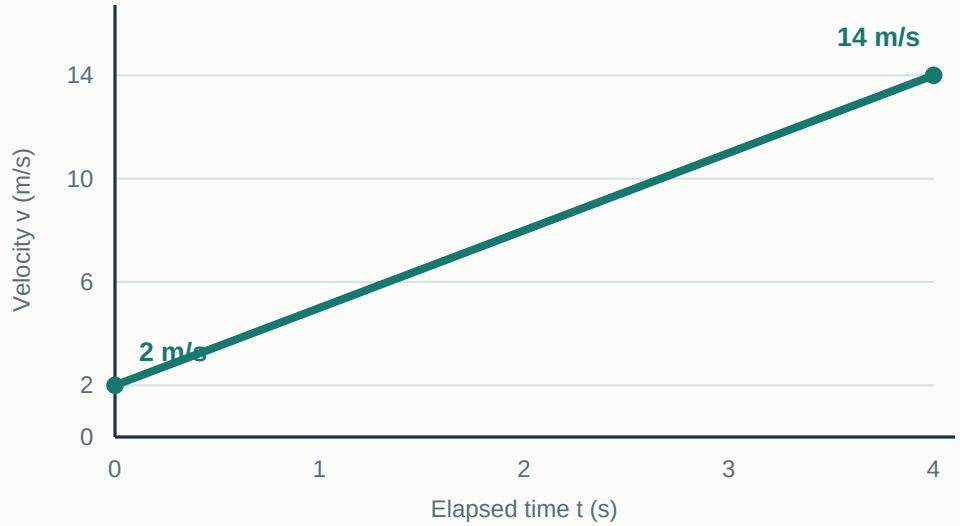
Start at a velocity of 2 m/s

Add 3 m/s of velocity every second

Watch the first 4 seconds of motion

$$v(t) = v_0 + a\,t$$

v0 = initial velocity a = acceleration t = time



The slope tells us how velocity changes; the line starts at the initial velocity.

Displacement is the area under velocity

The space below this velocity line is a trapezoid.
Its area tells us how far the cart moves forward.

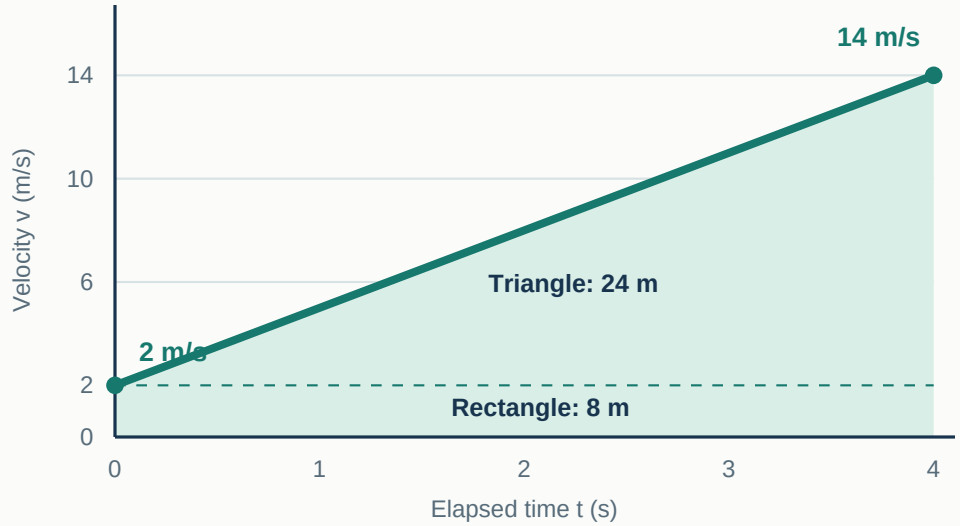
Average velocity is halfway between start and finish.

Multiply that average by the number of seconds.

$$\bar{v} = \frac{v_0 + v(t)}{2}$$

$$\Delta x = \bar{v}\, t$$

v-bar = average velocity delta x = displacement



The graph links how fast the cart moves to how its position changes.

Build the constant-acceleration equation

Put the velocity formula into the formula for the area.

Then multiply time into each part of the expression.

$$\Delta x = \frac{v_0 + (v_0 + a\,t)}{2}\,t$$

$$\Delta x = v_0\,t + \tfrac{1}{2}a\,t^2$$

The starting velocity gives the rectangle's area.

The extra velocity gives the triangle's area.

delta x = position change v0 = initial velocity a = acceleration t = elapsed time

Both parts give metres; use this formula only while acceleration stays constant.